

BUSA8031

Client Briefing Report – Assignment 2



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Executive Summary

This report examines population trends in Australia from 1996 to 2016, using census data given by the Australian Bureau of Statistics (ABS). The analysis employs Power BI dashboards to investigate changes in population size, age structure, gender distribution, and country of birth across regions.

The dataset covers all eight Australian states and territories, with extensive breakdowns by age, gender, year, and country of birth. Ten nations of birth were chosen for comparison: Australia, China, Thailand, Singapore, Germany, Ireland, Egypt, Fiji, New Zealand, and the United States. The selection process aims to reflect diverse continents while reducing regional bias.

The results indicate that New South Wales, Victoria, and Queensland experienced the majority of Australia's population expansion. Additionally, the largest percentages of residents are foreign-born in these states. Germany and Egypt had older, more settled populations, but China had the fastest-growing migrant population. The most frequent sources of youth migrants were Thailand and China.

In most groups, the gender split stayed equal. Although more than 91% of the population was born in Australia, the major eastern states had the greatest levels of cultural variety. This analysis supports planning in areas such as infrastructure, education, healthcare, and regional migration programs. It also demonstrates how Power BI can be used to extract distinct patterns from complicated data.

Introduction & Context

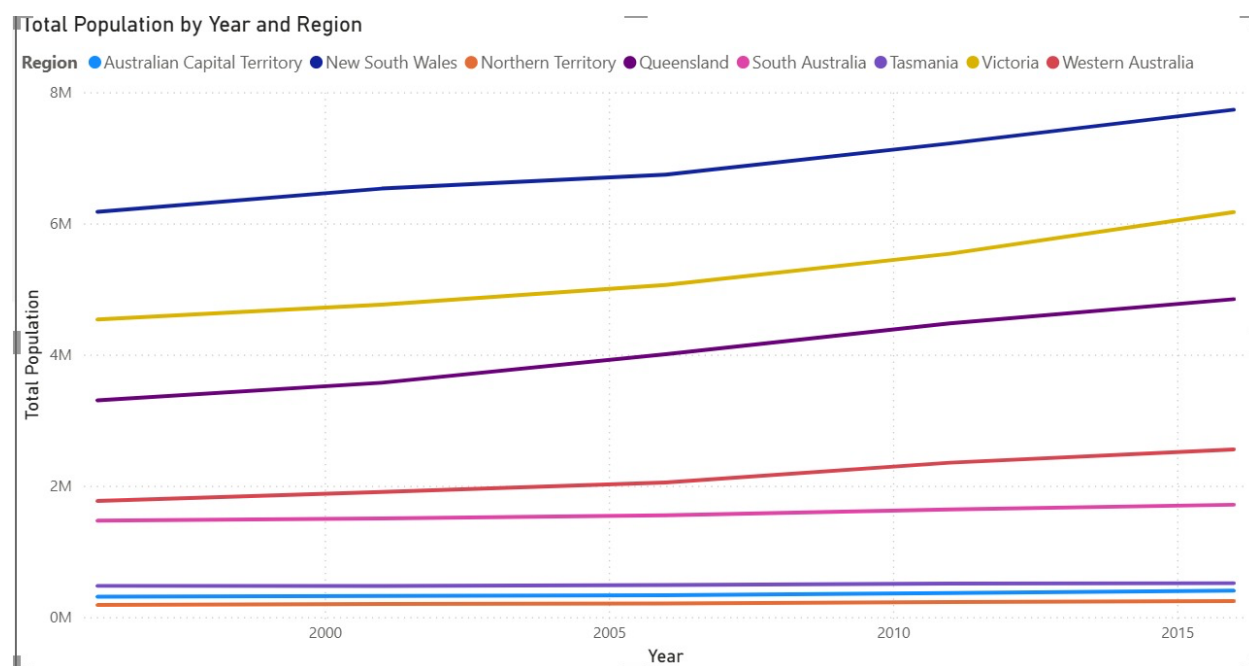
The goal of this research is to investigate and comprehend demographic trends in Australia using census data given by the Australian Bureau of Statistics. The study spans 1996 to 2016 and focuses on population growth, regional distribution, age structure, gender balance, and country of birth. Using Power BI dashboards, the report converts raw data into relevant graphics to aid policymaking, urban planning, and community service delivery.

The dataset used in this analysis contains population counts segmented by gender, age, year, region, and country of birth. It includes all eight Australian states and territories and records census intervals across five years. To deepen the analysis, ten countries of birth were selected to represent different global regions. These include Australia, China, Thailand, Singapore, Germany, Ireland, Egypt, Fiji, New Zealand, and the United States. The selection approach aimed to ensure diversity by including countries from Asia, Europe, Oceania, North America, and Africa. Countries were also randomly chosen to reduce bias and avoid over-representation from a single region, allowing for a more balanced view of Australia's migrant composition.

This analysis was undertaken to assist public sector planners in determining where population growth is occurring, which locations are the most diverse, and how age distribution influences service needs. It examines ten selected countries of birth over a 20-year period using ABS census data. The source file, "AUS_COB_ABS.csv," contains estimates of Australia's resident population broken down by state, age group, gender, and country of birth. The findings provide a comprehensive picture of demographic changes between 1996 and 2016 and are intended to assist policy decisions in healthcare, education, housing, and migration across Australian states and territories.

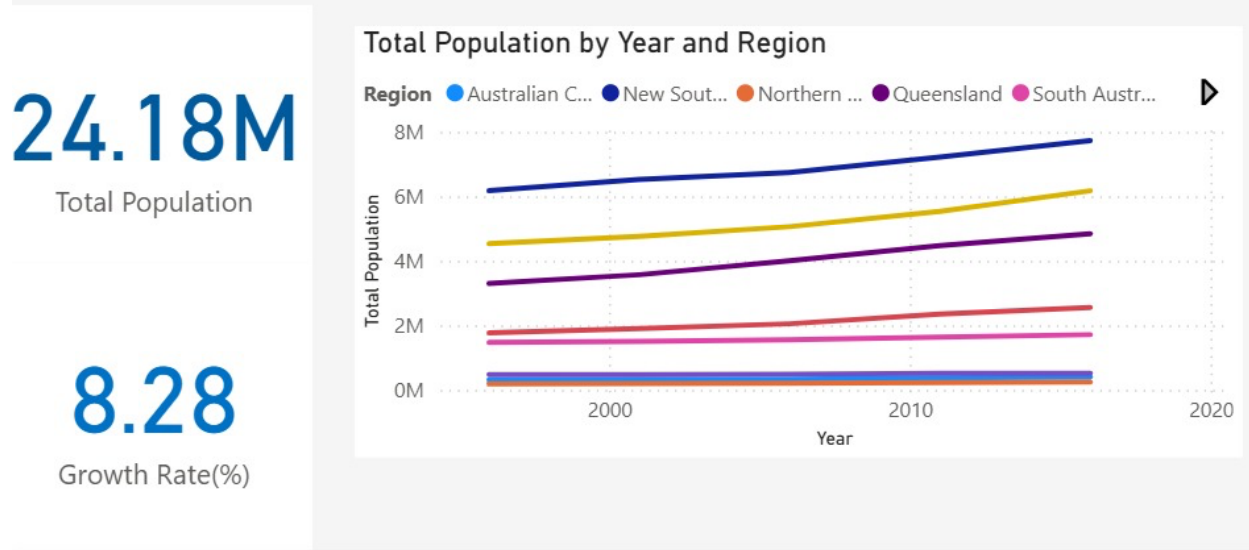
Dashboard 1: Total Population Overview by Region

From 1996 to 2016, this dashboard provides a thorough demographic overview of Australia's population dynamics and distribution throughout its states and territory. Four visualisations that highlight age segmentation, gender balance, geographical totals, and temporal growth provide the insights.



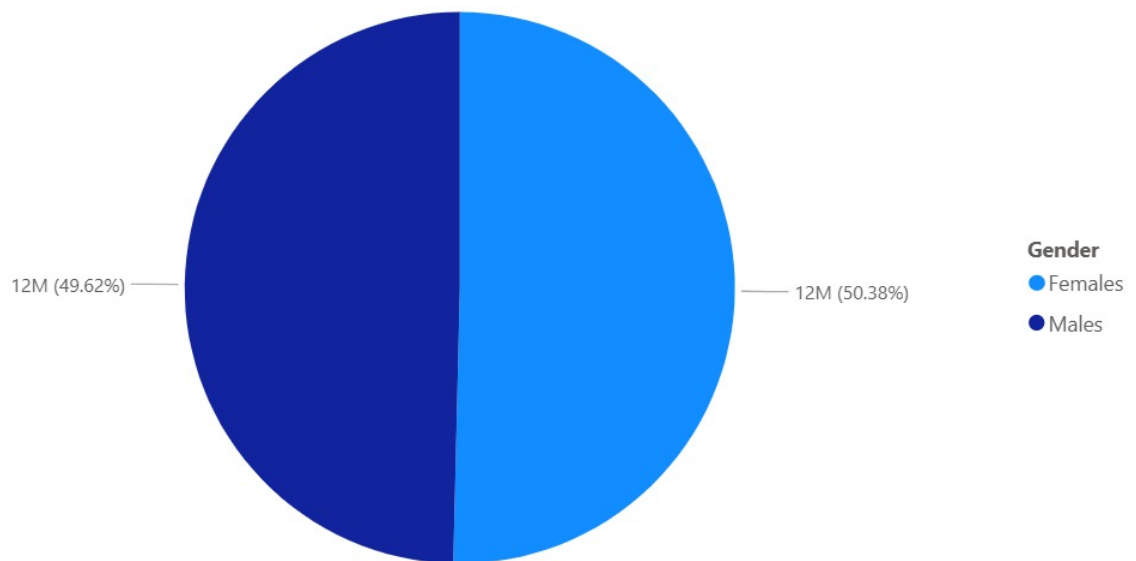
The line chart of total population by region and year shows a constant growing trend in all eight states and territories. Notably, New South Wales (NSW) and Victoria have maintained their status as the most populous areas, with both exceeding 6 million by 2016. Queensland has also seen fast growth after 2006, owing to rising domestic migration or birth rates. In contrast, places

such as the Northern Territory, Tasmania, and the Australian Capital Territory have comparatively low populations (under 1 million), implying reduced urbanization or limited migratory attraction. The visualization clearly shows the growing population disparity between highly developed and sparsely inhabited states.

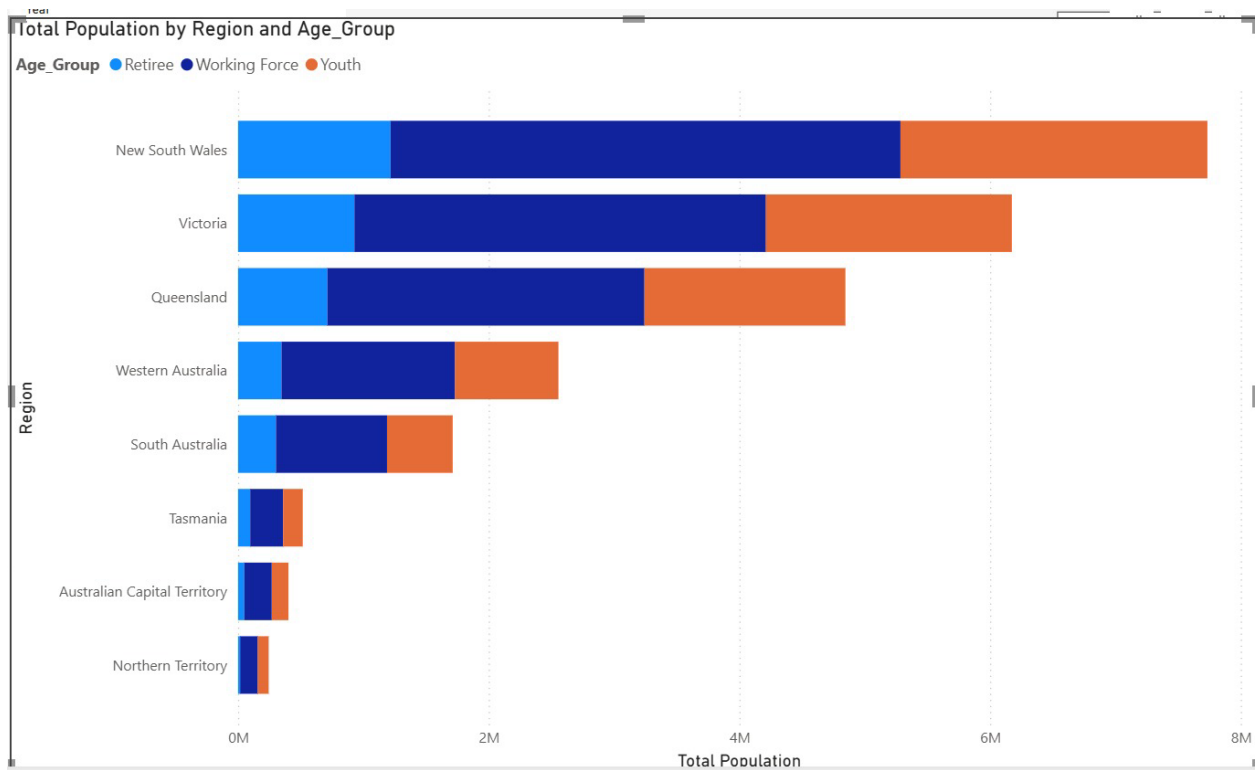


This tendency is supported by the growth rate card, which shows an 8.28% rise in national population between the earliest and most recent census years. This moderate but continuous trend predicts that demographic development will remain constant over the next two decades. It confirms Australia's constant expansion, highlighting the importance of long-term planning in healthcare, housing, and education, particularly in quickly increasing urban areas.

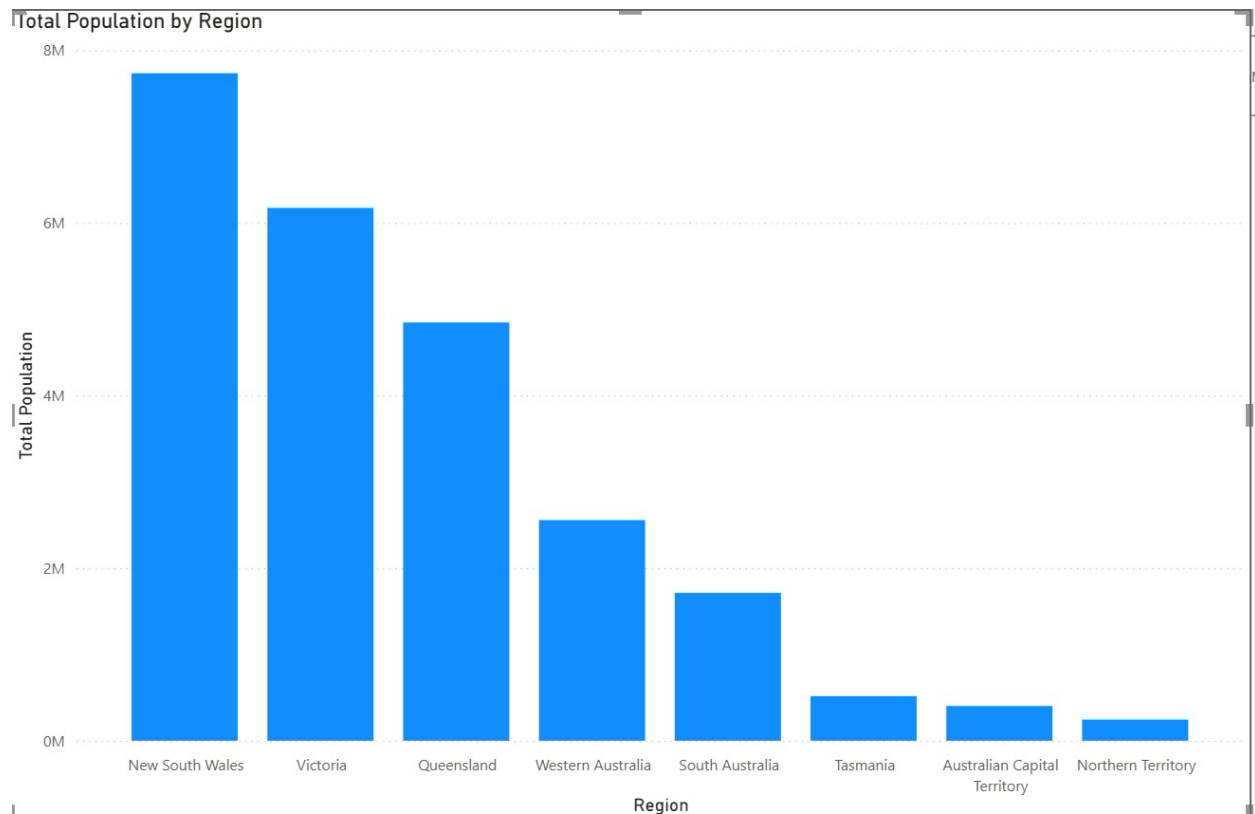
Total Population by Gender



The gender distribution pie chart shows an almost perfect population split, with males accounting for 50.38% and females for 49.62% in 2016. This parity is consistent with national census data and suggests a balanced demographic structure, decreasing the need for gender-specific policy actions. It also demonstrates the dataset's integrity by proving accurate modelling and aggregation via the gender dimension.



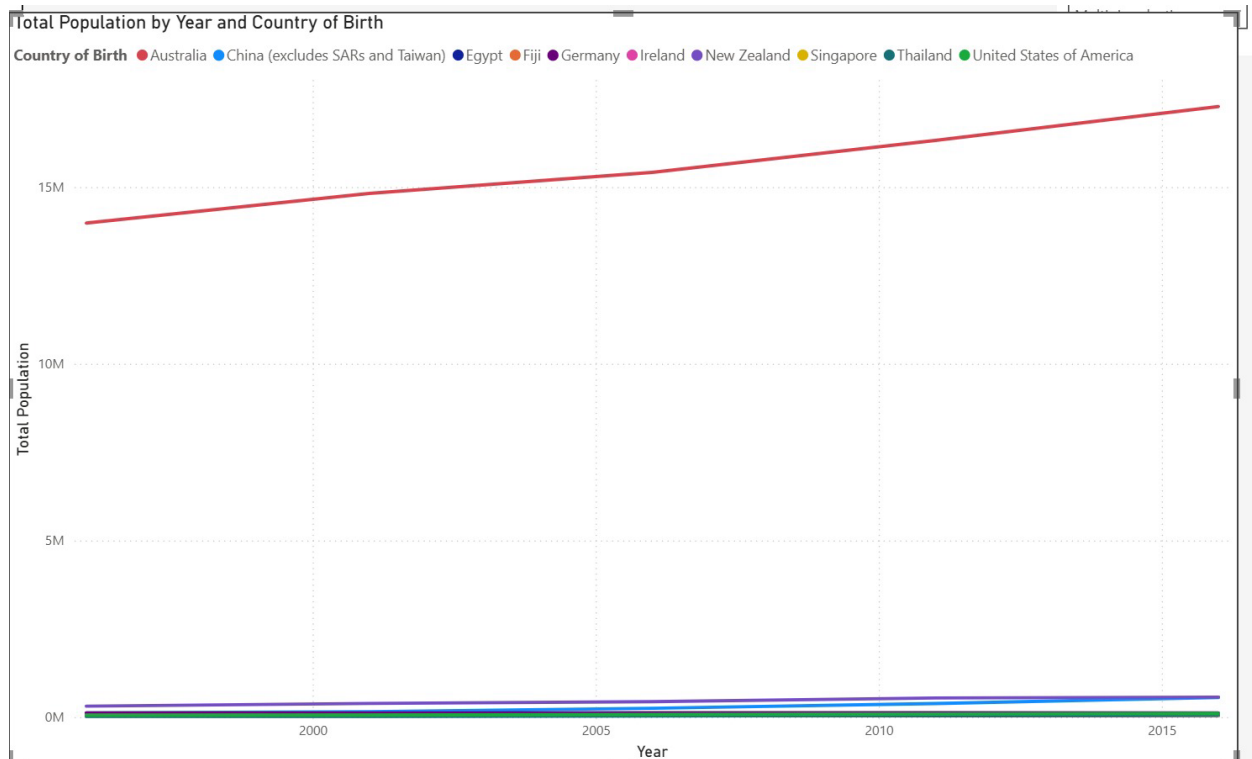
A stacked bar chart divides the population into three groups: youth (0-24), working age (25-64), and pensioners (65 and over), which are mapped against each region. The working-age population is dominant in all regions, particularly in New South Wales, Victoria, and Queensland. Youth populations are also common in these states, indicating a significant future labor force potential. South Australia and Tasmania have greater retiree populations, indicating potential need for improved aged care services.



Finally, a vertical bar chart allows for an easy comparison of overall population by region. It confirms the dominance of New South Wales, Victoria, and Queensland, followed by mid-tier areas (Western and South Australia) and less populous territories. This image reinforces the urban-rural divide by emphasizing crucial factors for infrastructure investment and regional development initiatives.

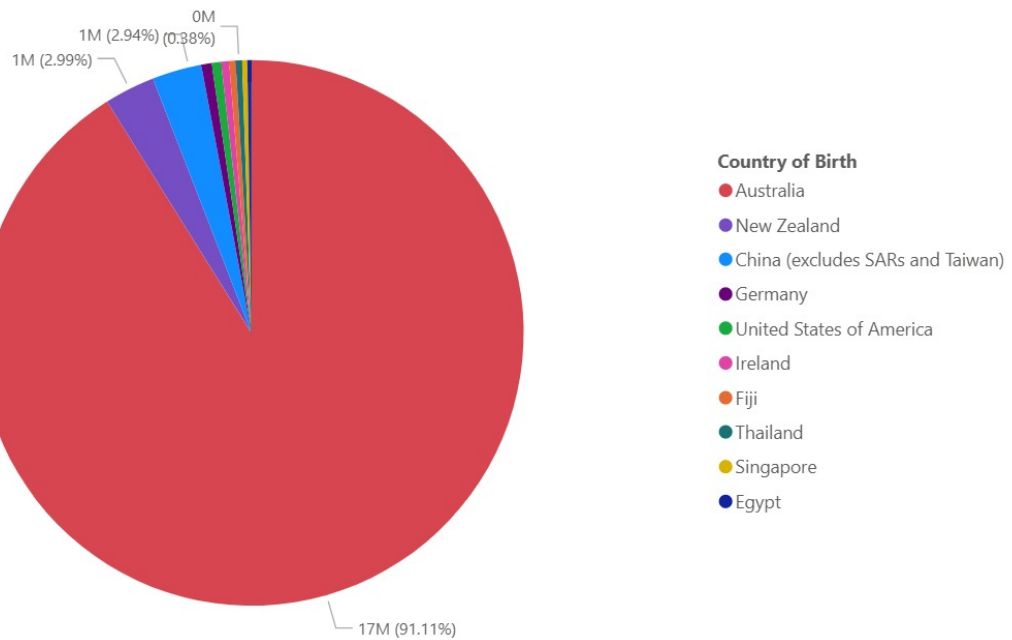
Dashboard 2: Selected Country of Birth (COB) Analysis

This dashboard provides a demographic picture of the population based on ten specific countries of birth, including Australia. The graphs compare population increase over time, age distribution, and gender composition.

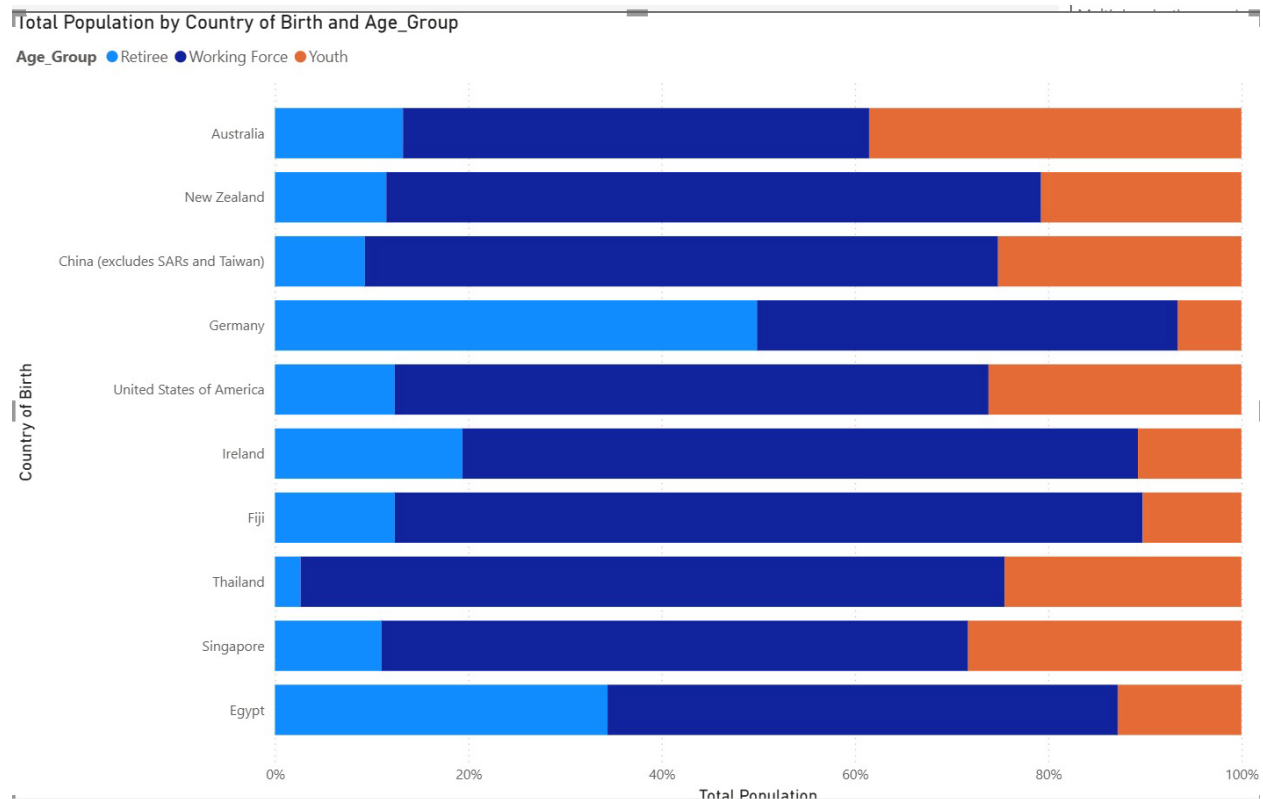


The first line chart, which covers five census years from 1996 to 2016, reveals that Australia has the largest population, with over 17 million by 2016. The proportion of New Zealanders born abroad is steadily increasing. China shows a rapid surge from less than 200 thousand in 2001 to nearly matching New Zealand by 2016. Other countries, such as Germany, Ireland, Egypt, and Fiji, have smaller but stable populations.

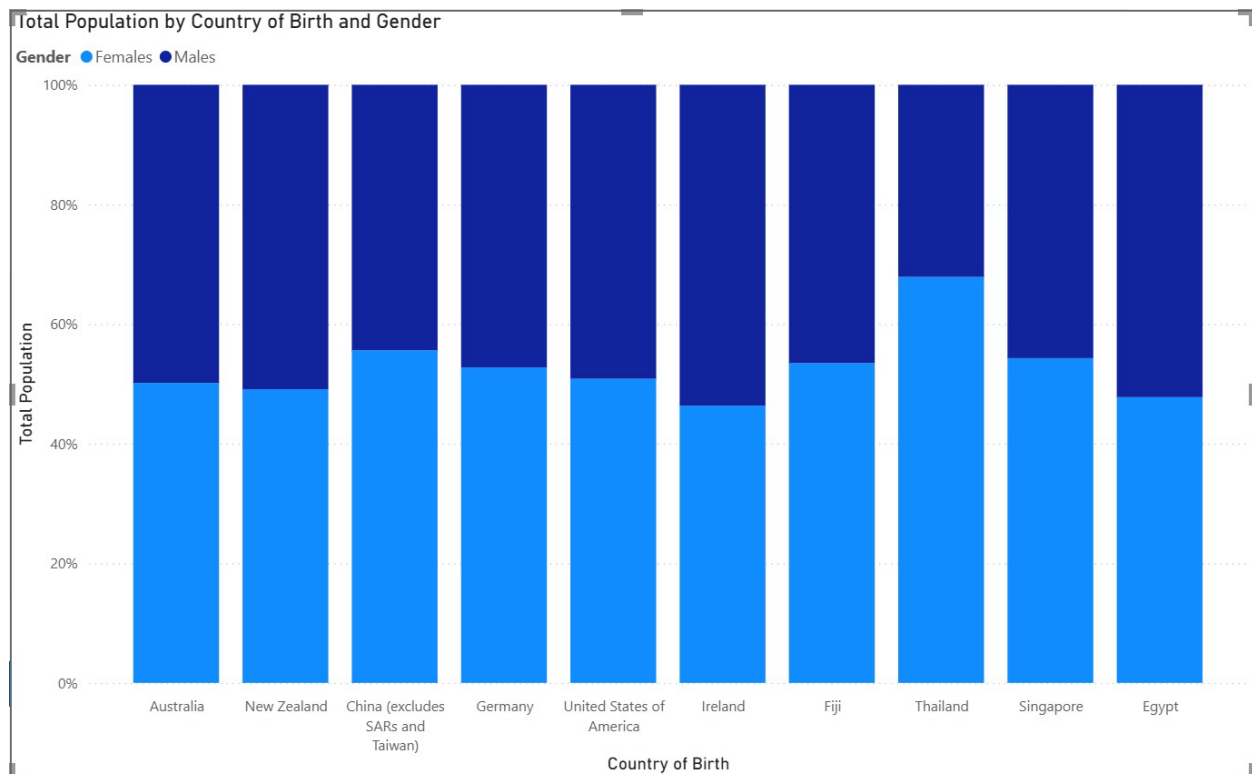
Total Population by Country of Birth



The second pie chart shows the population proportions by country of birth. Australian-born people account for 91.11 percent of the population. New Zealand gives 2.99 percent, followed by China at 2.91 percent. Countries like Fiji, Singapore, and Egypt all contribute less than 1%. This emphasizes the predominant native-born population and the smaller proportion of overseas-born communities.



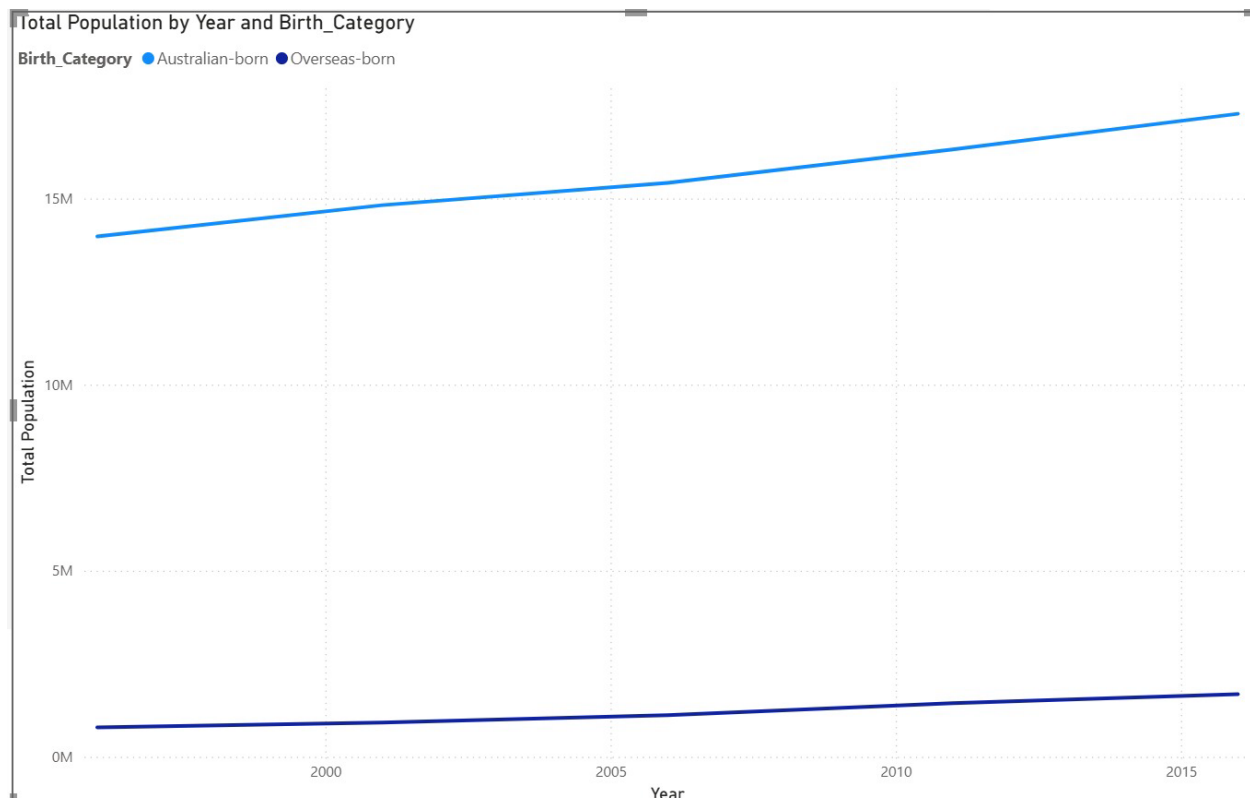
The third chart depicts each country's population divided by age group. Australia has a balanced distribution of young, working-age, and pensioner populations. China has the biggest proportion of youth, indicating student and early-career mobility. Germany and Egypt have higher retirement rates, which may reflect older and more stable cultures. Countries such as Fiji, Thailand, Singapore, and Ireland have more than 70% of their populations of working age. This shows that migration from these nations is mostly motivated by job and skilled visas.



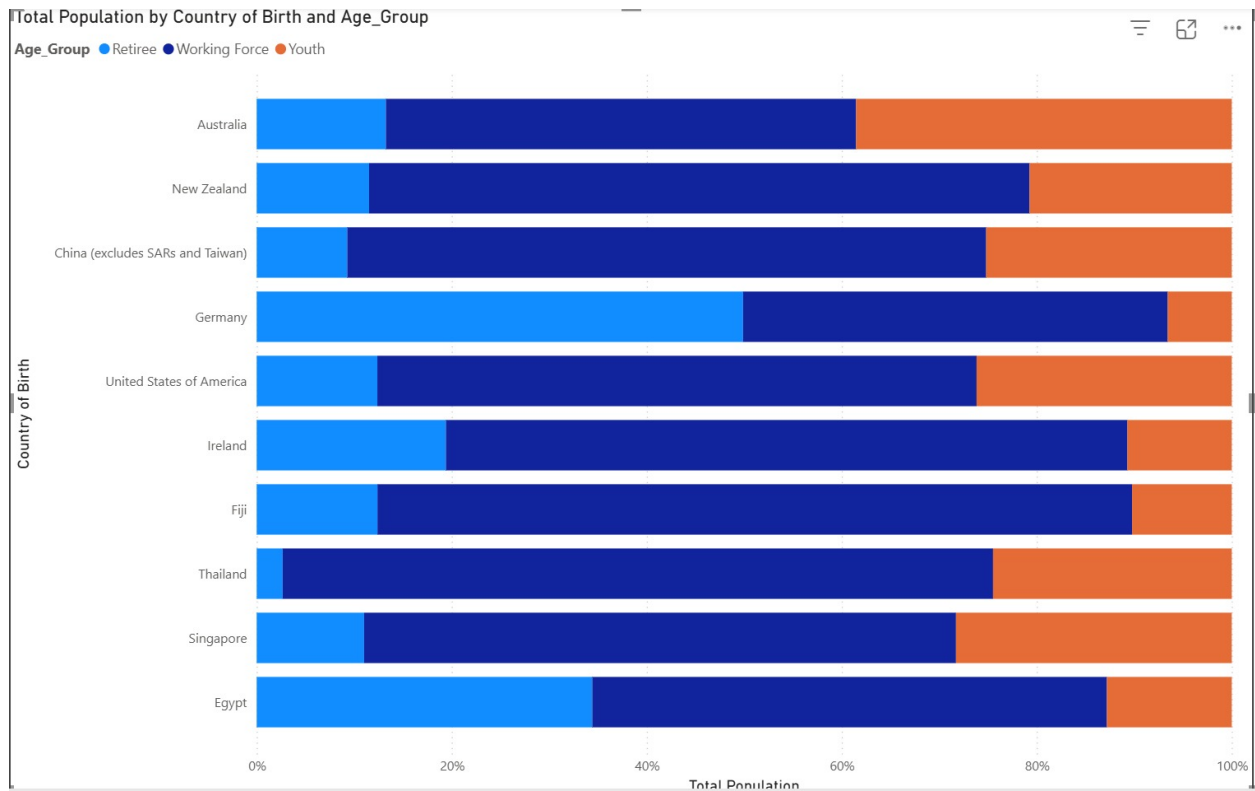
The final chart compares gender distribution by country of birth. Most countries, including Australia, New Zealand, and China, have an even distribution of males and females. Thailand has a female predominance, with more than 55% of the population being female. Egypt has a male majority. These patterns reflect several migratory motives. Female migration from Thailand may be tied to caregiving and education, but male migration from Egypt may be related to labor. The charts in this dashboard assist show how population size, age, and gender differ by country of birth. These findings aid in improved planning for services such as education, healthcare, employment, and community support.

Dashboard 3: Deeper Insights

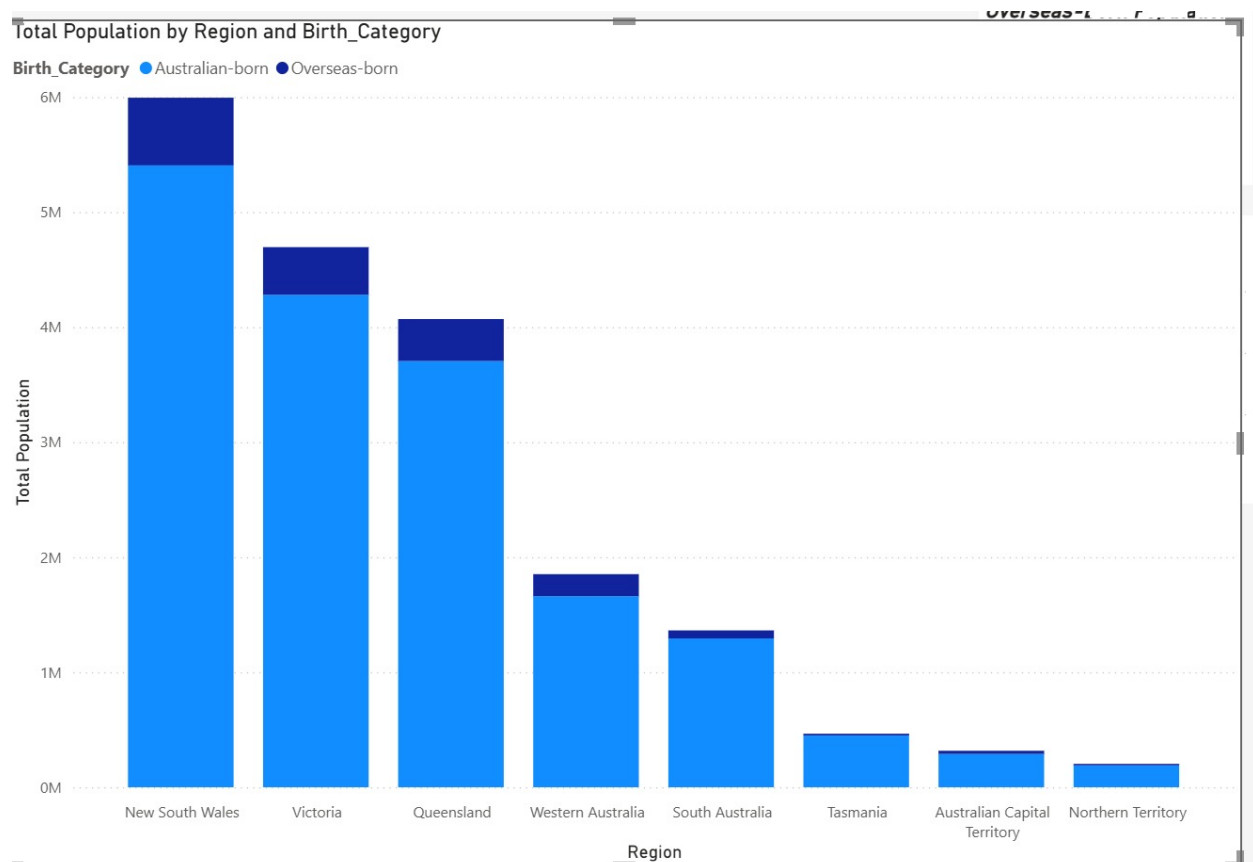
This dashboard contrasts Australian and overseas populations by year, age group, and area. It investigates how birth origin influences population structure and distribution throughout the country. The findings shed light on migration trends and demographic variance.



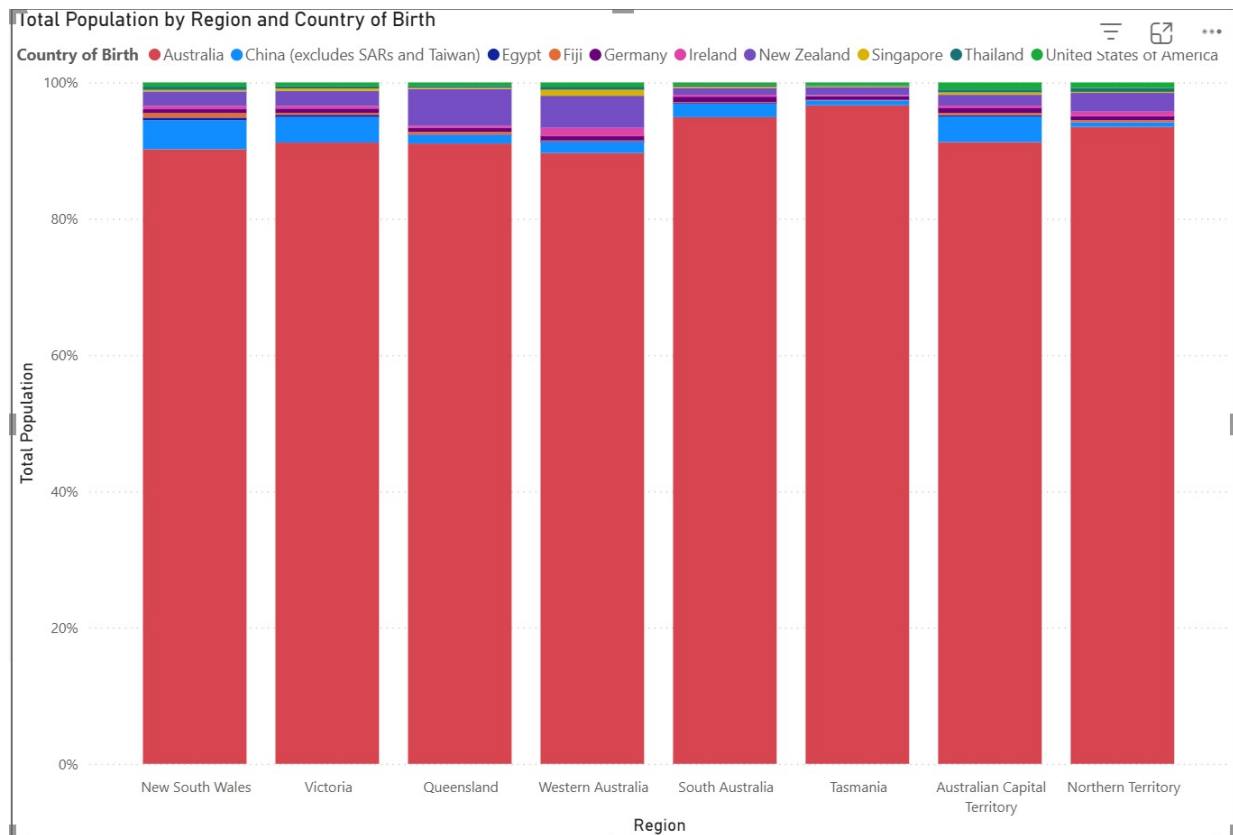
The first graph shows that the Australian-born population grew from around 13.5 million in 1996 to more than 17 million in 2016, with a strong majority across all years. However, the overseas-born population increased steadily, rising from approximately 3.5 million to more than 5.5 million during the same period. Despite its lower size, the overseas-born population is growing at a faster rate, reflecting an expanding migrant presence in Australia.



The second representation compares countries of birth by age group. Migrants from China and Thailand have the highest proportion of youth, indicating that education and early-career prospects are important drivers of migration from these countries. In contrast, Germany and Egypt have a higher share of retirees, indicating older, well-established societies. Singapore, Ireland, and Fiji have large working-age populations, with over 70% of individuals aged 25 to 64. Australia maintains a healthy balance throughout all three age categories, with a somewhat higher proportion of young and working-age people.



The third graph contrasts the population counts of Australians and those born overseas by region. New South Wales has the largest total population, exceeding 6 million, with a significant overseas component. Victoria and Queensland follow, with populations of over 4 million and 3.5 million, respectively. In contrast, places such as Tasmania, the Northern Territory, and the Australian Capital Territory have much smaller populations and lower migrant participation. This implies that most migrants settle in larger, more urbanized states.



The last chart depicts the proportionate diversity of each location. Australian-born residents account for more than 85% of the population in all states, with New South Wales and Victoria being the most varied. These states have a higher percentage of residents born in China, Fiji, and Germany. Western Australia and South Australia have modest diversity, whilst Tasmania and the Northern Territory have the lowest. This demonstrates that cultural diversity is concentrated in Australia's eastern and most populous regions.

These findings reveal that migrant patterns vary not only by volume but also by age and location. This highlights the need for region-specific planning, with more cultural services in urban states and migration incentives in less populated areas.

Key Findings & Recommendations

The data reveals that Australia's population has gradually expanded during the last two decades. The most populous states are New South Wales, Victoria, and Queensland, with each experiencing steady growth. These regions have a large proportion of working-age and young people, making them important places for labor and education planning. In comparison, South Australia and Tasmania have a higher proportion of retirees, which may boost demand for aged

care services.

Males and females make up an equal proportion of the population, both nationally and in most countries of birth. This balance suggests that gender-specific measures may not be urgently required. However, nations such as Thailand and Egypt have minor gender disparities, which could indicate distinct migration motives such as care work or labor responsibilities.

Majority of the Australia's population is Australian-born (91.11%). Among overseas-born populations, New Zealand and China contribute the largest shares. Migration from countries such as China and Thailand is driven by younger age groups, while Germany and Egypt have senior populations.

Migrants are mostly settling in major urban states. New South Wales and Victoria are the most culturally diverse regions. Smaller states like Tasmania and the Northern Territory show lower migrant representation.

These findings suggest that authorities should prioritize expanding infrastructure and services in high-growth metropolitan areas. Targeted migration initiatives could assist regional communities. Language help, employment routes, and elder care services should be adapted to each region's demographics and cultural profile.

Conclusion

This report used Power BI dashboards to analyze demographic data from Australia. It examined shifts over time and between states in terms of population size, age, gender, and country of birth. The dashboards assisted in displaying the demographics of the population, their differences by age and place of origin, and the changes in these trends between 1996 and 2016. The analysis found that most people live in New South Wales, Victoria, and Queensland. These states also have the largest overseas-born populations. Younger migrants tend to come from countries like China and Thailand, while older migrant groups are more common from Germany and Egypt. Exploring the data and identifying distinct patterns was made simple with Power BI. The visual aids made it easier to transform complicated data into insightful knowledge. Based on actual population changes, these dashboards help make better judgments about planning, policy, and service delivery.